

Lab 2: Microwave Optics

Objective

The objective of this lab is to experimentally investigate key wave properties of electromagnetic radiation in the microwave regime, including beam divergence, interference (standing waves), polarization response, and reflection/refraction behavior, using the Complete Microwave Optics System (WA-9316A) from Pasco. What follows will either be directly from the documentation or edited to make it clearer.

Caution

CAUTION: Under some circumstances, microwaves can interfere with electronic medical devices. If you use a pacemaker, or other electronic medical device, check with your doctor or the manufacturer to be certain that low power microwaves at a frequency of 10.525 GHz will not interfere with its operation.

CAUTION: The output power of the Microwave Transmitter is well within standard safety levels. Nevertheless, never look directly into the microwave horn at close range when the Transmitter is on.

Equipment

Gunn Diode Microwave Transmitter

The Microwave Transmitter, Figure 1a, provides 15 milliwatts (mW) of coherent, linearly polarized microwave output at a wavelength of 2.85 cm. The unit consists of a Gunn diode in a 10.525 gigahertz (GHz) resonant cavity, a microwave horn to direct the output, and an 18 cm stand to help reduce table top reflections. The Transmitter is powered directly from a standard 115 or 220/240 volt (V) alternating current (AC), 50/60 Hz outlet by using the included power adapter. Other features are a light emitting diode (LED) power-ON indicator and a rotational degree label on the back panel that allows easy measurement of the angle of polarization. The Gunn diode acts as a non-linear resistor that oscillates in the microwave band. The output is linearly polarized along the axis of the diode and the attached horn radiates a strong beam of microwave radiation centered along the axis of the horn.

Schottky Diode Microwave Receiver

The Microwave Receiver, Figure 1b, provides a meter reading that, for low amplitude signals, is approximately proportional to the intensity of the incident microwave signal. A microwave horn identical to that of the Transmitter's collects the microwave signal and channels it to a Schottky diode in a 10.525 GHz resonant cavity. The diode responds only to the component of a microwave signal that is polarized along the diode axis, producing a direct current (DC)

voltage that varies with the magnitude of the microwave signal. Special features of the Receiver include four amplification ranges—from one to thirty—with a variable sensitivity knob that allows fine tuning of the amplification in each range. The receiver is powered from a standard 115 or 220/240 VAC, 50/60 Hz outlet by using the included power adapter. and has an LED power-ON indicator. As with the Transmitter, an 18 cm high mount minimizes table top reflections, and a rotational degree label on the back panel allows convenient measurements of the polarization angle.

NOTE: The detector diode in the Receiver is a non-linear device. For sufficiently small signals, the detector output is approximately proportional to incident intensity, but due to diode nonlinearity this relationship is not strictly linear and may deviate at higher signal levels. This non-linearity will provide no problem in most experiments. However, it is important to realize that the meter reading is not directly proportional to either the electric field (E) or the intensity (I) of the incident microwave. Instead, it generally reflects some intermediate value.

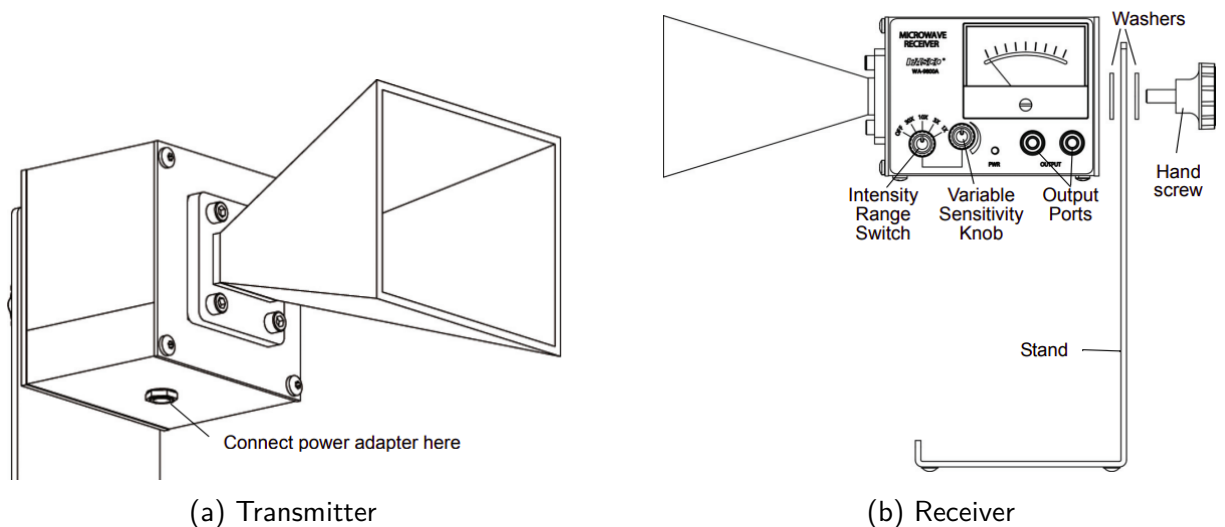


Figure 1: The PASCO Microwave Optics System transmitter and receiver.

Experiment

The stands for the transmitter and receiver have the effective transmission point, labeled with a T, and the effective receiver point, labeled with an R. Be sure to record your measurement uncertainty. Additionally, be sure there are no objects near the transmitter, receiver, or on the table between them - keep the area very tidy to minimize stray reflections and multipath interference.

Angular Beam Spread

Be sure the transmitter and receiver are orientated such that the slits within the horns are vertical. Position the Transmitter so the output surface of the horn is centered directly over the center of the Degree Plate of the Goniometer arm, Figure 2. With the Receiver directly facing the

Transmitter and as far back on the Goniometer arm as possible, adjust the Receiver controls for a meter reading of 1.0. **NOTE:** Do not change the sensitivity once this is done. Now measure the signal at angles between $\pm 90^\circ$ from directly facing the transmitter. Increase the number of measurements where you are able to detect signal (ie every 2° vs 10° when the signal is below detection threshold). Define the beam width as the full-width at half-maximum (FWHM) of the detected signal. Plot this data with a polar plot.

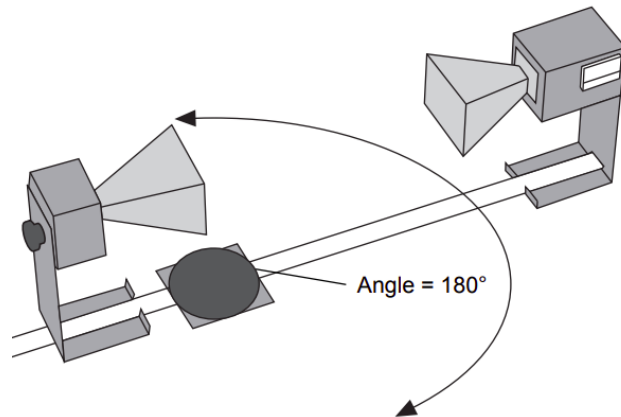


Figure 2: Experimental setup to determine the spread of the beam.

Standing Waves

Return the Receiver to being aligned with the Transmitter (ie angle is 0°). Place the Transmitter at the 50.0-cm mark and the Receiver at the 85.0-cm mark. Adjust the sensitivity to 30 and the variable sensitivity to a reading of 1.0 at this location. Gently move the Receiver 2-mm farther away from the Transmitter (ie to 85.2-cm), make sure the Transmitter and the angle hasn't changed and then record the signal reading. Continue in this manner until you have 3-4 cycles of max to min and back. Standing waves arise due to reflection of microwaves from the receiver and transmitter, producing interference between forward and reflected waves.

Note: the maximum likely won't return to 1.0 but it will clearly be a maximum. Plot this data along with plots of $\cos(kx)$ and $\cos^2(kx)$ to Determine the wavelength from the spacing between adjacent maxima (or minima), which corresponds to $\lambda/2$. If the detector responds linearly to the electric field, the signal should follow $\cos(kx)$; if it responds to intensity, it should follow $\cos^2(kx)$. Compare your measured wavelength to $\lambda = c/f$ using $f = 10.525$ GHz.

Polarization

Return the Receiver to being aligned with the Transmitter (ie angle is 0°). Place the Transmitter at the 10.0-cm mark and the Receiver at the 100.0-cm mark. Adjust the sensitivity to 30 and the variable sensitivity to a reading of 1.0 at this location. The microwave radiation from the Transmitter is linearly polarized along the Transmitter diode axis (i.e., as the radiation propagates through space, its electric field remains aligned with the axis of the diode). If the Transmitter diode were aligned vertically, the electric field of the transmitted wave would be vertically polarized, as

shown in Figure 3a. If the detector diode were at an angle θ to the Transmitter diode, as shown in Figure 3b, it would only detect the component of the incident electric field that was aligned along its axis.

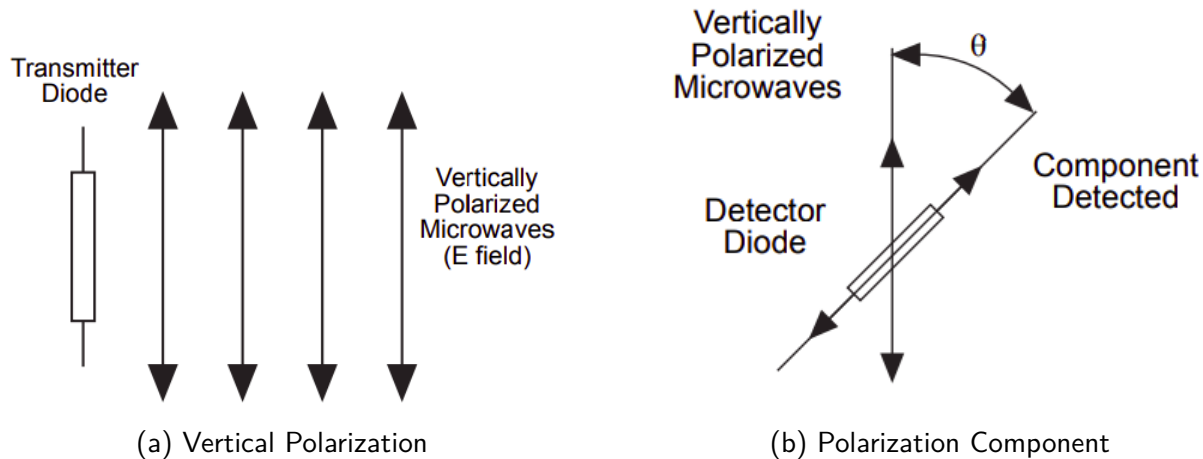


Figure 3: Polarization of Microwave Radiation

The math works out the same if we rotate the Transmitter instead of the Receiver (plus it is easier to read the signal). Starting with the Transmitter and Receiver vertically aligned take signal readings every 5° for a full 180° . Plot this data along with plots of $\cos(\theta)$ and $\cos^2(\theta)$. If the detector measures electric field amplitude, the signal is proportional to $\cos(\theta)$; if it measures intensity, the signal is proportional to $\cos^2(\theta)$. Which model better fits your data, and what does this imply about the detector response?

Reflection

Refer to Figure 4 for the setup. Place the Transmitter at 25-cm on the fixed arm of the Goniometer and the Receiver at 100-cm. Be sure to adjust the Transmitter and Receiver to the same polarity; the horns should have the same orientation. The reflector is a thin metal plate adhered to a magnetic stand with a hole for the bolt on the Goniometer. The angle between the incident wave from the Transmitter and a line normal to the plane of the Reflector is called the Angle of Incidence (see Figure 4). Adjust the Rotating Component Holder so that the Angle of Incidence equals 45° . Without moving the Transmitter or the Reflector, rotate the movable arm of the Goniometer until the meter reading is a maximum. The angle between the axis of the Receiver horn and a line normal to the plane of the Reflector is called the Angle of Reflection. Set the scale to 10 and adjust the variable sensitivity to get a reading of 1 at this angle. For Angles of Incidence from 20° to 80° in steps of 10° verify that the angle of reflection equals the angle of incidence within experimental uncertainty. Record your values in a table. NOTE: At various angle settings the Receiver will detect both the reflected wave and the wave coming directly from the Transmitter, thus giving misleading results.

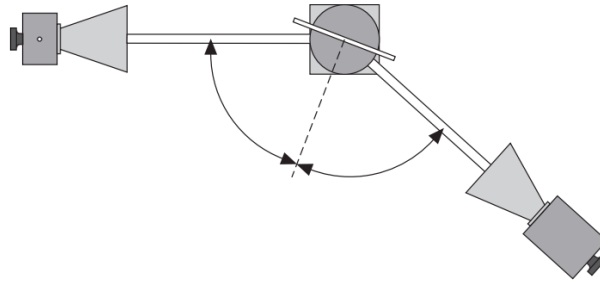


Figure 4: Experimental setup to check the Law of Reflection.

Refraction

An electromagnetic wave usually travels in a straight line. As it crosses a boundary between two different media, however, the direction of propagation of the wave changes. This change in direction is called **refraction**, and is summarized by a mathematical relationship known as the **Law of Refraction** (otherwise known as **Snell's Law**):

$$n_1 \sin \theta_1 = n_2 \sin \theta_2;$$

where θ_1 is the angle between the direction of propagation of the incident wave and the normal to the boundary between the two media, and θ_2 is the corresponding angle for the refracted wave, see Figure 5.

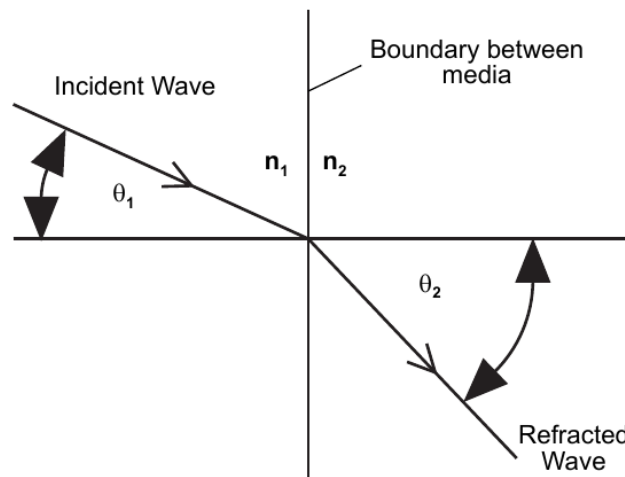


Figure 5: Illustration to show geometry for Snell's Law.

Every material can be described by a number n , called its **Index of Refraction**. This number indicates the ratio between the speed of electromagnetic waves in vacuum and the speed of electromagnetic waves in the material, also called the medium. In general, the media on either side of the boundary will have different indices of refraction (and the difference between wave velocities that this implies) which causes “bending”, or refraction of a wave as it crosses the

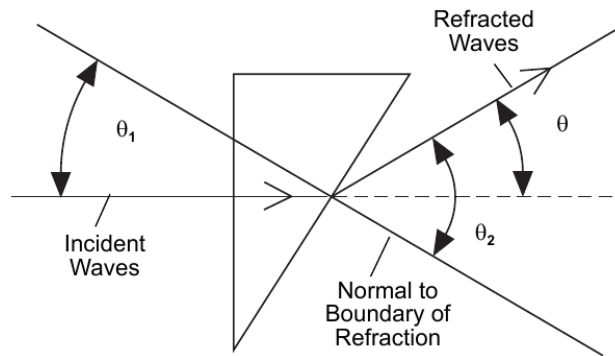


Figure 7: Illustration to show relevant angles in the refraction experiment.

boundary between two distinct media. Note: The index of refraction is generally dependent on the frequency.

Before you begin use any method you like to determine the small angle of the styrofoam prism; this is θ_1 . Then arrange the equipment as shown in Figure 6 with the transmitter at 10-cm mark of the fixed arm of the Goniometer and the receiver at the 100-cm mark of the movable Goniometer arm. Set the receiver to a multiplier of 3. Orient the transmitter and receiver such that the slits are horizontal. Fill the styrofoam prism with styrene beads. Gently rotate the free arm of the Goniometer to find the angle, θ , of maximum signal.

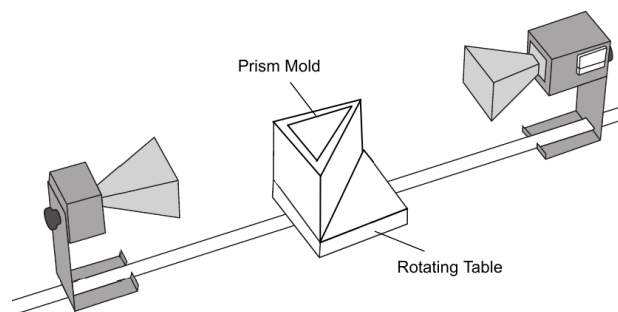


Figure 6: Illustration to show equipment layout.

The angles are tricky but Figure 7 helps to clarify things. The incoming microwaves enter the prism at the flat face at normal incidence. A fraction of the microwave's amplitude is transmitted into the prism, n_1 , and propagates to the hypotenuse of the prism such that they strike the surface at an angle, θ_1 . These waves reflect and refract at the boundary with the refracted waves exiting into air, $n_2 = 1.0$, at an angle θ_2 with respect to the normal. From the geometry determine a relationship between θ_1 , θ_2 , and θ .

Use Snell's Law and your measured angles to determine the index of refraction of the styrene pellets (the styrofoam is essentially transparent at this frequency).

Report

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